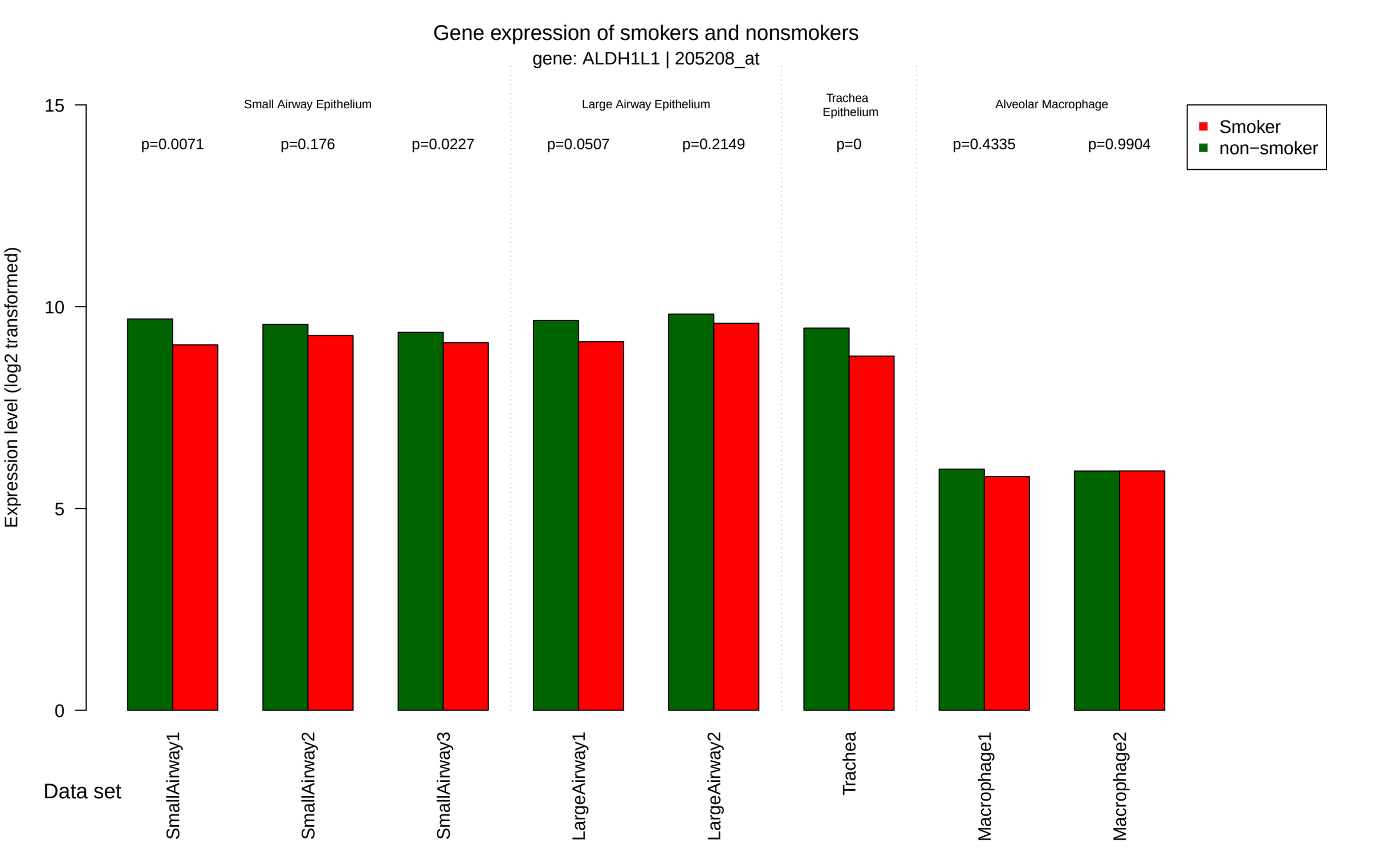
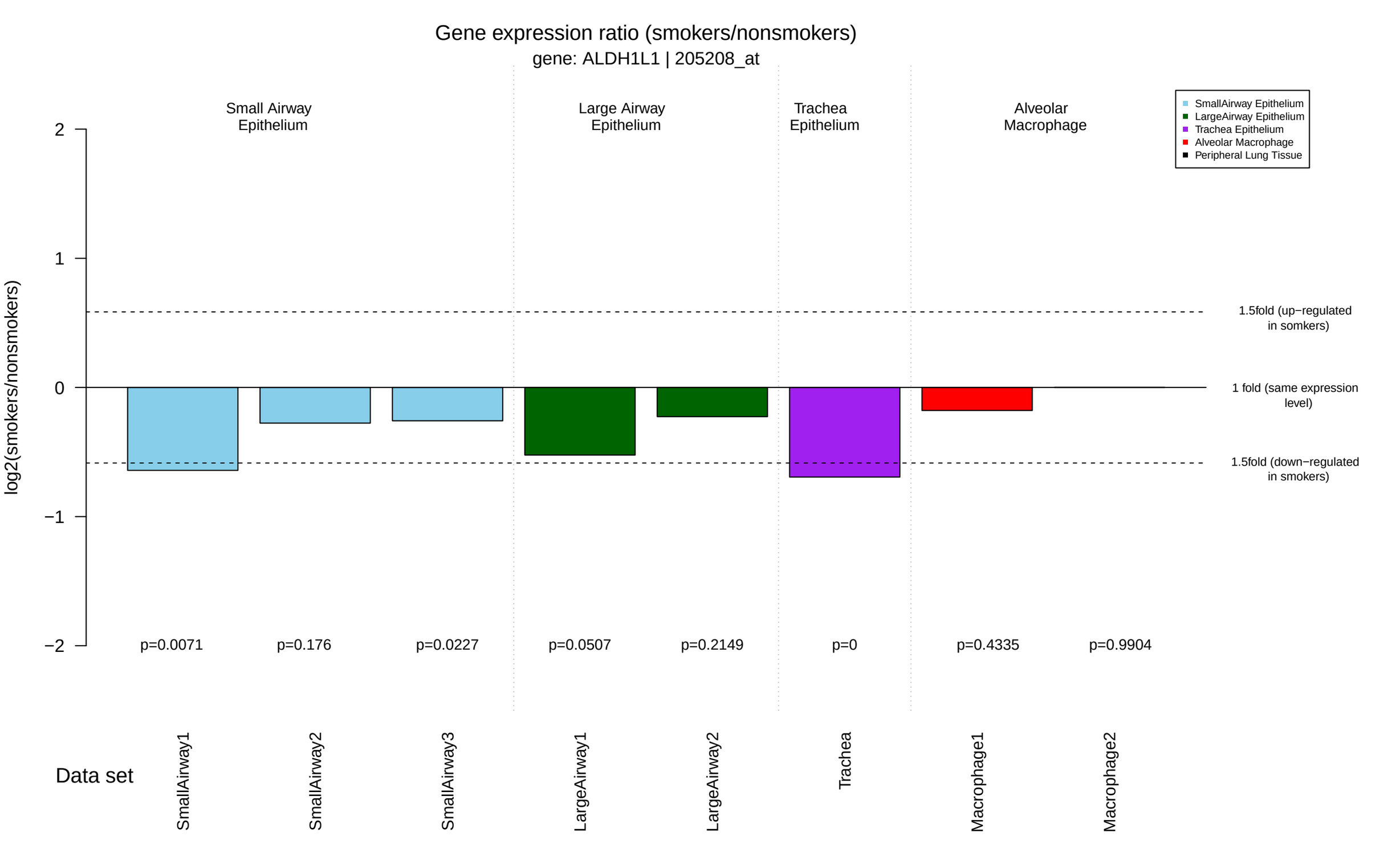
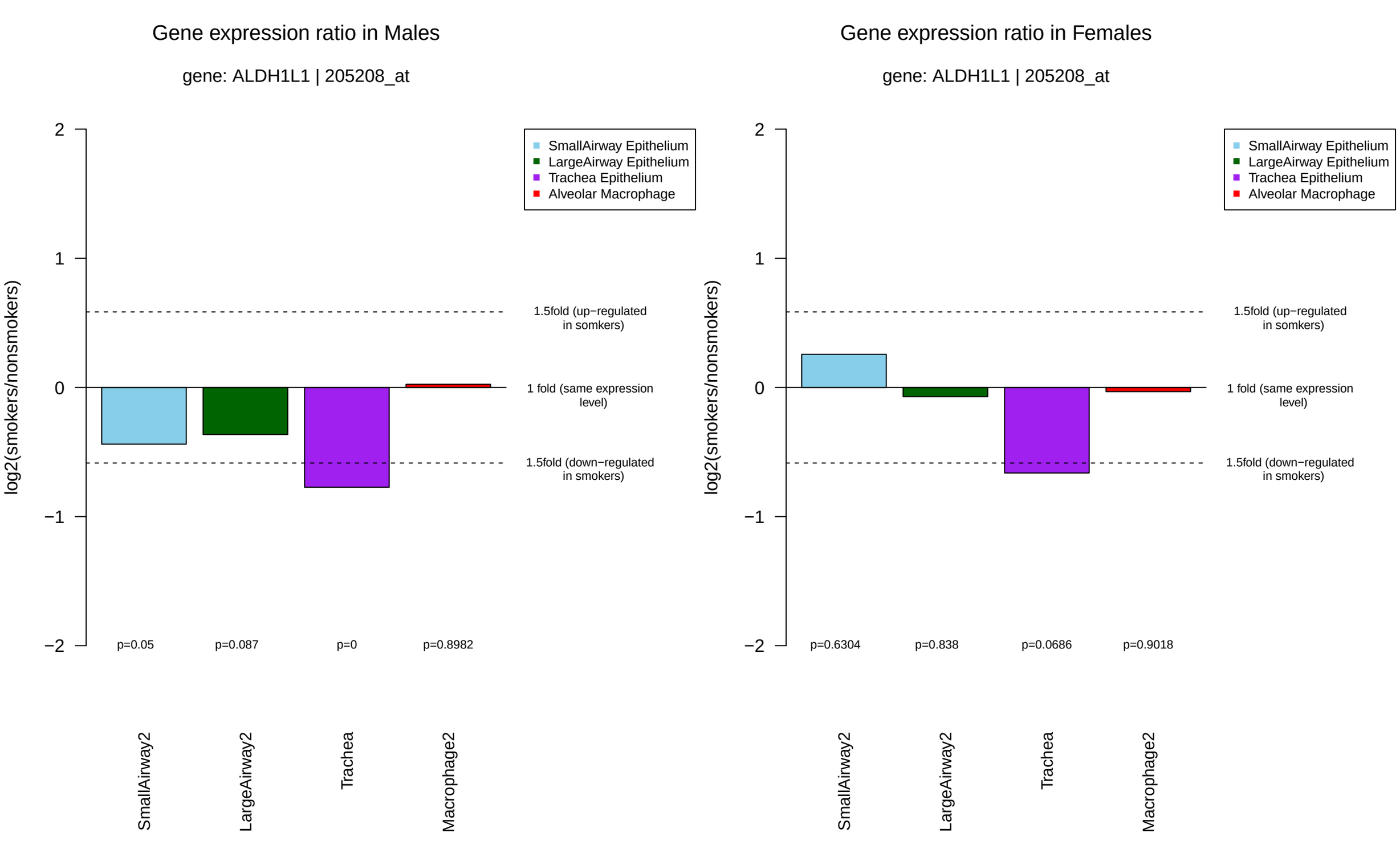
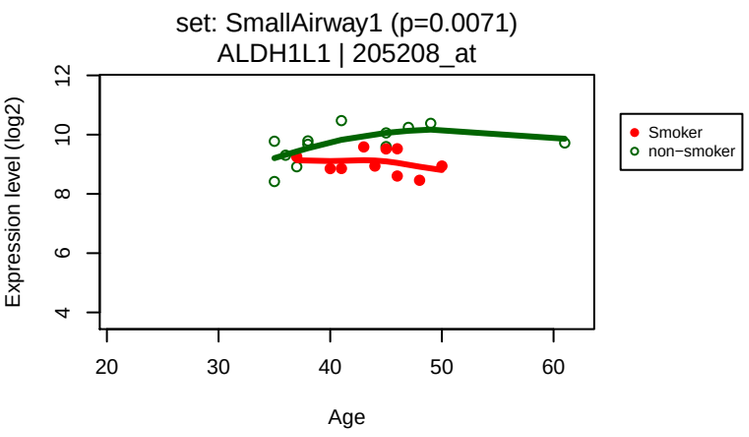
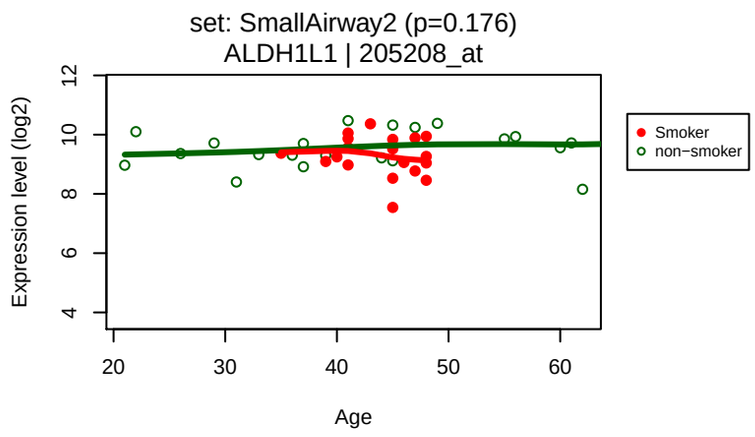
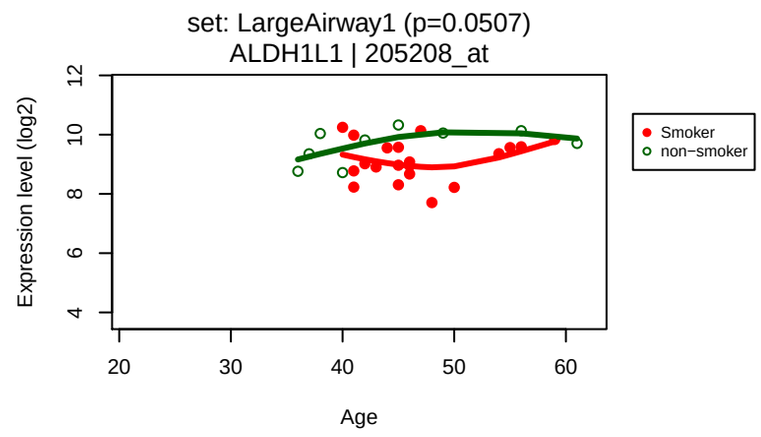

Gene expression of smokers and nonsmokers
gene: ALDH1L1 | 205208_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0071
p=0.176
p=0.0227
p=0.0507
p=0.2149
p=0
p=0.4335
p=0.9904
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

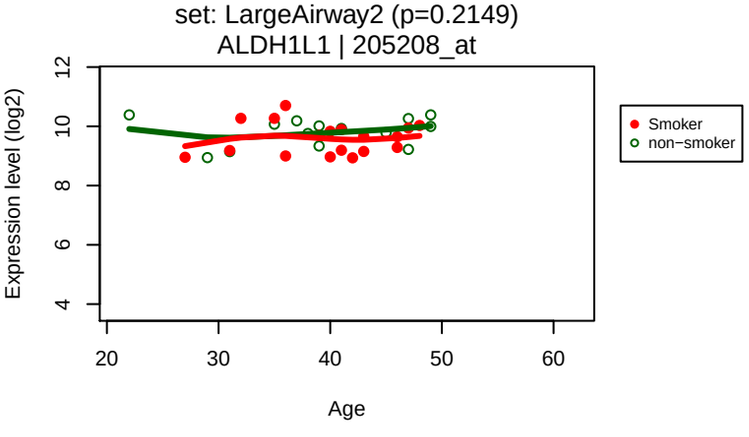

Gene expression ratio (smokers/nonsmokers)
gene: ALDH1L1 | 205208_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0071
p=0.176
p=0.0227
p=0.0507
p=0.2149
p=0
p=0.4335
p=0.9904
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

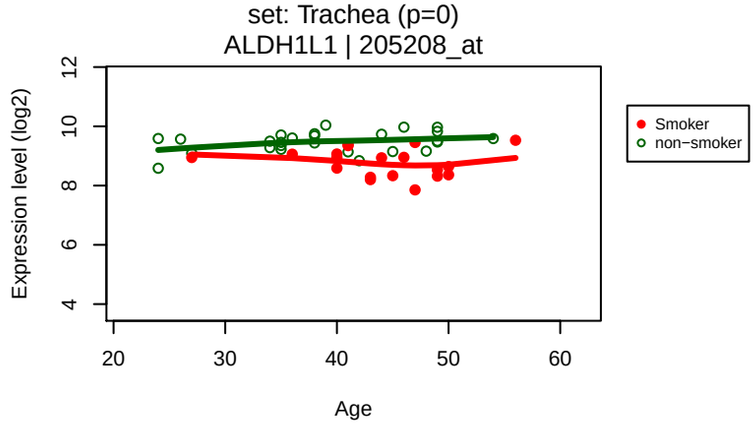

Gene expression ratio in Males
gene: ALDH1L1 | 205208_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.05
p=0.087
p=0
p=0.8982
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH1L1 | 205208_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6304
p=0.838
p=0.0686
p=0.9018
SmallAirway2
LargeAirway2
Trachea
Macrophage2

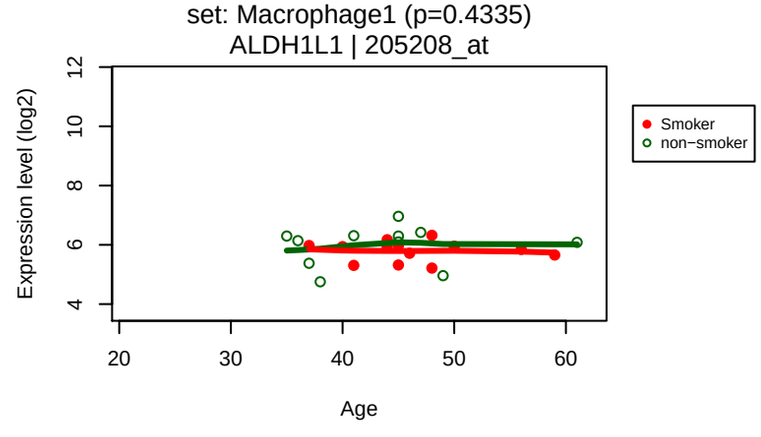

set: SmallAirway1 (p=0.0071)
ALDH1L1 | 205208_at
Expression level (log2)
Age
Smoker
non−smoker

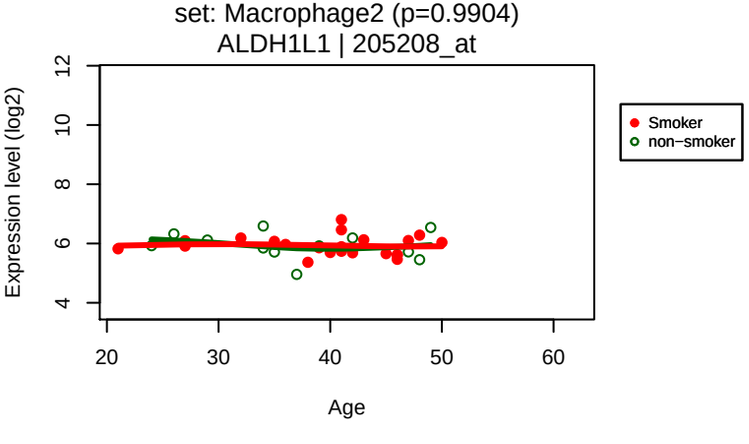

set: SmallAirway2 (p=0.176)
ALDH1L1 | 205208_at
Expression level (log2)
Age
Smoker
non−smoker

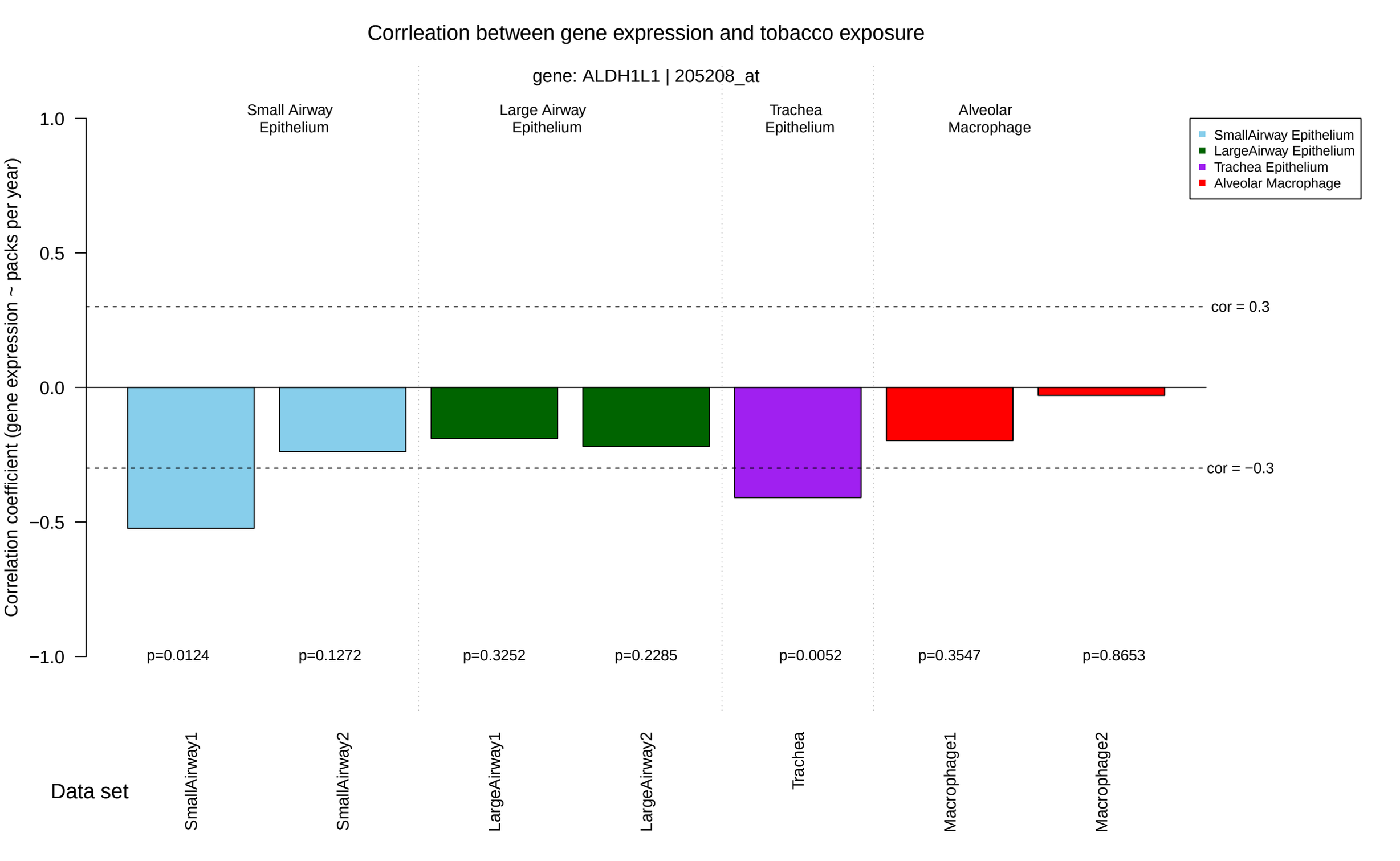

set: LargeAirway1 (p=0.0507)
ALDH1L1 | 205208_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2149)
ALDH1L1 | 205208_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
ALDH1L1 | 205208_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4335)
ALDH1L1 | 205208_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9904)
ALDH1L1 | 205208_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH1L1 | 205208_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0124
p=0.1272
p=0.3252
p=0.2285
p=0.0052
p=0.3547
p=0.8653
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2